| Question | Answer | Marks | Partial Marks | Guidance |
|---|---|---|---|---|
| 7(d) | FT reading at 14 weeks from *their* straight line of best fit with positive gradient | **1** | | Allow for intention of a straight line/ dashed line<br>Tolerance ± 0.05 of *their* reading at 14 |
| 8(a) | 2.1 | **2** | **B1** for 4.2 [cm]<br>or **M1** for answer *their* written measurement × 0.5 correctly evaluated | Accept answers 2 to 2.2 [km]<br>For B1 accept 4 to 4.4 [cm]<br>For M1 *their* measurement must lie between 4 cm and 5 cm |
| 8(b) | Correct construction for position of *L* with intersecting arcs 8.8 cm from *A* and 6.6 cm from *B* | **3** | **B2** for correct position of *L* indicated with no/incorrect arcs<br>or for completely correct with intersecting arcs but *L* to the west of *A* and *B*<br><br>or **B1** for 8.8 cm from *A* or 6.6 cm from *B* soi<br><br>If 0 scored, **SC1** for fully correct with arcs but distances reversed 6.6 cm from *A* and 8.8 cm from *B*<br>or for position of *L* to the east with intersecting arcs 4.4 cm from *A* and 3.3 cm from *B* | Tolerance ± 2 mm for all marks – use ruler tools to measure<br>Accept any clear unambiguous indication of *L*, e.g. a dot, the intersection of arcs<br><br>Accept e.g. an arc 8.8 cm from *A* |
| 9(a) | $7.09 \times 10^{-3}$ | **1** | | Condone trailing zeros in (a) and (b) |
| 9(b) | $1.6 \times 10^{9}$ | **2** | **B1** for $16 \times 10^{8}$ oe | |
| 10(a)(i) | Triangle drawn at (2, 0), (1, –1) and (3, –3) | **2** | **B1** for reflection in $y = k$<br>or for reflection in $x = 1$ | Mark intention, accept freehand, use overlay as a guide in part (a)(i) and (ii) |
| 10(a)(ii) | Triangle drawn at (–4, 3), (–5, 4) and (–3, 6) | **2** | **B1** for translation by $\begin{pmatrix} -6 \\ k \end{pmatrix}$ or by $\begin{pmatrix} k \\ 1 \end{pmatrix}$ | |

| Question | Answer | Marks | Partial Marks | Guidance |
|---|---|---|---|---|
| 10(b) | Rotation<br><br>90° anticlockwise oe<br><br>[centre] (3, –1) | **3** | **B1** for each | More than one transformation scores zero<br>Not turn<br><br>Accept 90, 270 clockwise, – 270, 90 counterclockwise, 90 a/c<br><br>Condone omission of brackets in coords, accept '<u>position</u> vector' $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$<br>but not vector $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$<br>If the centre has been described as a vector (word or bracket notation) then do not treat as additional transformation |
| 11(a) | 1842.59 | **2** | **M1** for $1500 \times \left(1+\frac{4.2}{100}\right)^5$ oe | Mark at most accurate<br>For 2 marks accept 1840, 1842, 1843, 1842.5, 1842.6, 1842.59…<br>M1 may be done in stages and method can be implied by values given to 3 sf or better from previous stage[s]<br>$1500 \times (1 + 4.2\%)^5$ is not enough for M1<br>Method spoiled if<br>e.g. $1500 + 1500 \times \left(1+\frac{4.2}{100}\right)^5$<br>but allow M1 if $1500 \times \left(1+\frac{4.2}{100}\right)^5 - 1500$<br>M1 implied by answer 343, 342.5, 342.6, 342.59… |